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Homemade fusion reactor

He built a fusion reactor in his garage when he was 14, and now gives a TED talk on how he did it <http://dgit.in/HomeFusRe>

Enhancing your Gaming Experience on Linux

Yes gaming on Linux is a reality. Make it a reality that you enjoy with these tweaks.

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2014 has been a great year for Gaming on Linux, and 2015 can only improve on it. This year saw the release of AAA games such as Civilization V, XCOM: Enemy Unknown, Tropico 5, Borderlands 2, and Borderlands: The Pre-Sequel, and also high profile indie releases such as Wasteland 2, Planetary Annihilation, Dreamfall Chapters, and even Broken Age. There are of course hundreds of other games smaller titles that were either ported to, or released for Linux this year. Linux Gaming is definitely a 'thing' now, a enough of a thing to make a popular web store like Gog.com change its mind and start supporting the platform.

Even before the very idea of this Linux gaming revolution was but a blip in anyone's radar, there were credible arguments for Linux as a gaming platform. Linux has been for a long time, the gaming platform that gamers deserved, but didn't really need. You are justified, of course, in asking why we think so.

It all comes down in the end, to what makes a platform good for gaming. Gaming is a very specialized kind of computer activity. It taxes the computer in ways that other software don't, to the extent that people often have specialised hardware (graphics cards) that are of greatest benefit only while gaming.

When playing a game, you would hope to have as little interference between the Game itself, and the bare metal of your hardware. As few as possible other software running. It's why a console like the PlayStation 3 could manage for so long with a now laughable 512MB RAM.

Linux happens to have a huge advantage in reducing the gap between the game and the underlying hardware. It is endlessly configurable to the point that it can run efficiently on both tiny watches, and supercomputers that take up a room or few.

```

root@kali:~# nvidia-smi
Mon Nov 17 09:04:02 2014
+-----+
| NVIDIA-SMI 343.22     Driver Version: 343.22 |
+-----+-----+
| GPU Name      Persistence-M| Bus-Id        | Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap| Memory-Usage  | GPU-Util  Compute M. |
+-----+-----+-----+
| 0. GeForce GTX 480 SE  Off  | 0000:01:00:0  | N/A    | N/A                  |
| 41%   50C   P8     N/A /  N/A | 815M / 1024M | N/A     | Default              |
+-----+-----+-----+
| Compute processes: |
| GPU  PID  Process name | GPU Memory |
|-----|-----|-----|
| 0          Not Supported |             |
+-----+-----+-----+
root@kali:~#

```

You can control the formatting and output of this command to a large extent.

Linux can go the distance, and it only needs someone to take charge.

'Hardcore' gamers often tweak their system to an extreme, disabling every unnecessary service, or even downloading sketchy versions of unbloated Windows (often going by names such as MicroXP) or dubious legality. They invest in expensive and elaborate cooling systems, just to push FPS from 80 to 82.

Linux of course encourages you can go as deep as you want and remove whatever you don't need. Go ahead change the maximum CPUs count in the kernel config from

512 to 4 and save KILOBYTES of RAM! Or, do something more useful maybe? The point is you have the option, if you want to you can recompile software for your specific configuration to boost performance, and if even you don't do all this yourself, someone will and they will release a gaming-specific distro just for you. Like what Valve is doing with SteamOS.

With that said, let's get to the crux of the matter. You've listened to our advice and switched to Linux (kudos!) or you were already using Linux (and we take

retroactive credit for that) and now you want to do some bloat-busting, and generally spruce up your Linux gaming experience.

Bustin' Bloat

Turn off unnecessary services

Linux is not immune to the curse of unnecessary services. The ones we list can be turned off to improve performance, at least while gaming. Some

of these are 'core' services that provide basic features, others might be specific to distro, desktop environment, or other software installed.

Here are a few services that you can do without while gaming:

- **vmware / virtualbox / libvirt** Unless you have one of those supercomputer-that-fills-a-room-or-few chances are you won't be running a virtual machine while playing games.
- **avahi** Mostly used for discovery of network services like shared media libraries and such.